

JONATHAN YEUNG

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WORK EXPERIENCE

Trellis Energy

San Francisco, CA

Associate Software Engineer

Jan 2025 – present

- Resolved database bottlenecks in a high-volume natural gas trading platform, rewriting Oracle SQL queries with eager and batch fetching to eliminate N+1 patterns, and **cutting page load times 71% (18.2s → 5.3s) and query time 65x (13.08s → 202ms)**
- Diagnosed and resolved production incidents, including a concurrency lock issue that blocked users from submitting time-sensitive operator confirmations, resolved by batching record lookups to **cut transaction times by 66% (2.06s → 703ms)**
- Built new full-stack platform features based on customer needs** for natural gas trading, trader credit-limit authorization, and pipeline capacity and interconnect quantity validation
- Drove Spring Boot 4 migrations across 7 backend services** and built a Claude skill to automate framework upgrades via OpenRewrite, resolving dependency and namespace conflicts and verifying with build, test, and startup checks

ARRIS

Cupertino, CA

Lead Software Engineer Intern

Jun 2024 – Sep 2024

- Selected by the CFO to **develop a financial accounting system for complex revenue calculations** to align sales/finance operations
- Implemented an ASC 606 compliant algorithm for calculating monthly recurring revenues, **improving reporting accuracy by 35%**
- Created a data ingest pipeline with integrity checks to **validate and link 3,000+ invoice records to sales and pricing contracts**

CodeLab

Davis, CA

Software Engineer Intern

Oct 2022 – Jun 2023

- Developed an algorithm to auto-generate anomaly-detection queries, with **MongoDB caching for improved retrieval performance**
- Built an event scheduler to scan Apache Pinot databases and **send critical AWS SES alerts based on changes to baseline data**
- Created an analytics dashboard with React and Spring REST APIs to **allow non-technical users to track and manage real-time alerts**

Innopage

Hong Kong

Blockchain Developer Intern

Jun 2022 – Sep 2022

- Developed the payment infrastructure system for a blockchain-based marketplace, **supporting \$750K in NFT sales within 24 hours**
- Implemented an autoscaling batch process for transactions based on network demand, **reducing transaction gas fees by 25%**
- Deployed a smart contract system with Solidity to automate sales, **facilitating 5,000 transactions / hour** on the Ethereum blockchain

Hong Kong Science and Technology Park

Hong Kong

Machine Learning Engineer Intern

Jun 2020 – Sep 2020

- Developed a deep learning vision model for the HK Hospital Authority to **track 100+ surgical tools for surgeries across 42 hospitals**
- Built an automated data tagging pipeline to classify 700GB+ of raw data, **reducing manual labeling times by 80%**
- Deployed on-premises AI infrastructure, utilizing Nvidia Quadro GPUs with CUDA-X for **ML training across 50 virtual machines**

EDUCATION

University of California, Davis

Davis, CA

B.S. Computer Science

June 2025

Double Minor in Statistics and Technology Management

Coursework: Vehicle and Transportation Systems, Machine Learning, Computer Networks, Operating Systems, Regression Analysis

PROJECTS

Level 1 Adaptive Cruise Control Simulation: Simulated adaptive vehicle distance control in Godot, using LiDAR sensor data and a 60Hz PID feedback control loop to manage engine power based on the target speed and leading vehicle distance

High-Performance Network Protocol: Designed a custom congestion control algorithm with transmission window resizing based on real-time packet delay, achieving 2.68x better throughput-to-latency compared to existing TCP protocols in a simulated Docker network

Cancer Cell Classification: Trained and deployed a model to classify tumors as malignant or benign with 96% accuracy, using Python for dataset preparation, cross-model evaluation (for logistic regression, SVM, and neural network models), and hyperparameter tuning

SKILLS

Languages: Python, Java, SQL, JavaScript/TypeScript, C, C++, R, HTML, CSS

Technologies: Docker, Node.js, Oracle DB, MongoDB, Linux, NumPy/Pandas, Flask, React, Spring Boot, Hibernate/JPA, Git, Figma

Additional Interests: Autocross (Hyundai Elantra N), Marathons, Golfing, Cycling